

Kothamrita Chakraborty

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LinkedIn | GitHub | Portfolio

Technical Skills

Languages	Python, C++, JavaScript, SQL
AI/ML	Machine Learning, Computer Vision, LLM Applications
Frameworks	LangChain, OpenAI API, Gemini API, Vector Databases, Prompt Engineering, React
Libraries	Pandas, NumPy, Matplotlib, OpenCV, Scikit-learn, Cloudflare Workers
Tools	Git, GitHub, VS Code

Projects

Margdarshak AI

Traffic Prediction & Decision-Support System

IIT Bombay Techfest Finalist

[Source Code](#)

Tech Stack: Python, Pandas, Scikit-learn, Matplotlib

- Developed an ML-based traffic congestion prediction system using real-time traffic data collected through global APIs.
- Engineered time-series features and trained predictive models achieving approximately 85% prediction accuracy for congestion forecasting.
- Selected as an IIT Bombay Techfest Finalist for building a decision-support framework for route optimization and traffic flow management.

Glaucat

AI-Based Glaucoma & Cataract Risk Detection

[Source Code](#)

Tech Stack: Python, TensorFlow, OpenCV

- Developed a TensorFlow-based computer vision system for glaucoma and cataract risk assessment using real-time eye tracking and visual response analysis.
- Implemented camera-based screening workflows that monitored eye movement patterns through interactive clinic-inspired exercises and games.
- Designed an automated screening pipeline to support accessible and early-stage eye disease detection using laptop and mobile device cameras.

MPLADS AI

AI-Powered Monitoring & Analytics Platform

[Source Code](#)

Tech Stack: HTML, CSS, JavaScript, TypeScript, Cloudflare Workers

- Contributed to the development of an AI-powered platform for monitoring MPLADS fund utilization and project implementation, aimed at improving transparency and accountability.
- Worked on database-related development and backend workflows using Cloudflare Workers to support the handling of project data for monitoring expenditure, work progress, and implementation status.
- Collaborated on version control and code management using Git and GitHub, supporting team coordination and integration of project components.

Experience

Indian Statistical Institute | Research Intern

Aug 2025 – Sep 2025

- Performed exploratory data analysis on research datasets using Python and data visualization libraries.
- Developed visualizations to identify trends, patterns, and actionable insights from complex datasets.
- Prepared analytical reports and documentation supporting research-oriented workflows.

Infosys Springboard | Web Development Intern

Nov 2025 – Jan 2026

- Developed AI-powered multi-agent systems using LangChain, Gemini, OpenAI APIs, and Vector Databases for research, summarization, email assistance, and task automation.
- Designed and implemented specialized agents including Research Assistant, Summarizer, Email Assistant, and Task Management Assistant.
- Integrated LLM-powered workflows into web applications using React, Bootstrap, and Prompt Engineering.

Education

B.Tech in Computer Science and Engineering	2024 – 2028 (Expected)
Meghnad Saha Institute of Technology	
CGPA: 7.3 (Till Sem 4)	
Class XII (Higher Secondary)	2024
Aaloke Bharti Model School	
Class X (Secondary)	2022
St. Teresa’s Secondary School	
Percentage: 92%	

Achievements & Certifications

- IIT Bombay Techfest Finalist – Margdarshak AI
- Qualified International Computer Science Competition (ICSC)
- NASA Space Apps Challenge Participant
- Programming Fundamentals Using Python – Infosys Springboard